

Closed-form 2nd-order Data Shapley as a Spurious-Correlation Attenuator

Where it Works and Where it Doesn't

Anonymous Authors | ICLR 2026 Workshop | code: github.com/anon/fairds

+13.0 pp

OOD test_worst vs Vanilla
($p=5\times 10^{-5}$, 10 seeds)

+4.7 pp

2nd-order isolation
(vs Fairds-1, $p=0.033$)

+55.3 pp

at strong spurious $p=0.99$
(phase transition, $p=3\times 10^{-5}$)

1.89×10^{-8}

Mann-Whitney p
(majority $\phi <$ minority ϕ)

Background

Empirical risk minimization on *spuriously-correlated* data latches onto majority shortcuts.

Mainstream remedy: **bi-level meta-learning** (Ren 2018; FORML) – implicit-differentiation heavy.

Wang et al. 2024 derived a *closed-form* 2nd-order In-Run Shapley estimator – but used only **post-hoc for filtering**.

Question: can the same closed-form formulas serve as an *in-training* reweighter? Does the 2nd-order cross-term contribute a *stand-alone* benefit?

Contributions

- Adapt closed-form In-Run Shapley as a 5-step in-training reweighter.
- RMS-rescale the cross-term** – prevents divergence under strong spurious bias.
- Statistically isolate a **2nd-order benefit** on Colored MNIST (+4.7 pp, $p=0.033$).
- Honestly characterize the **regime boundary** (works on from-scratch / small-model; fails on pretrained ResNet).

References & Code

Wang et al. (2024). *Data Shapley in One Training Run*. ICLR.

Ren et al. (2018). *Learning to Reweight Examples*. ICML.

Liu et al. (2021). *Just Train Twice*. ICML.

Sagawa et al. (2020). *GroupDRO*. ICLR.

Arjovsky et al. (2019). *IRM*. arXiv:1907.02893.

Code: github.com/anon/fairds

Method: Fairds

Per-step on minibatch $\mathcal{B}_t$ with anchor $\mathcal{D}_{\text{val}}$:

$$\phi_i^{(2)} = \langle g_i, g_{\text{val}} \rangle - \alpha \cdot \langle g_i, H_{\text{val}} g_{\text{val}} \rangle$$

► g_i via `torch.func.vmap(grad)` (single graph).

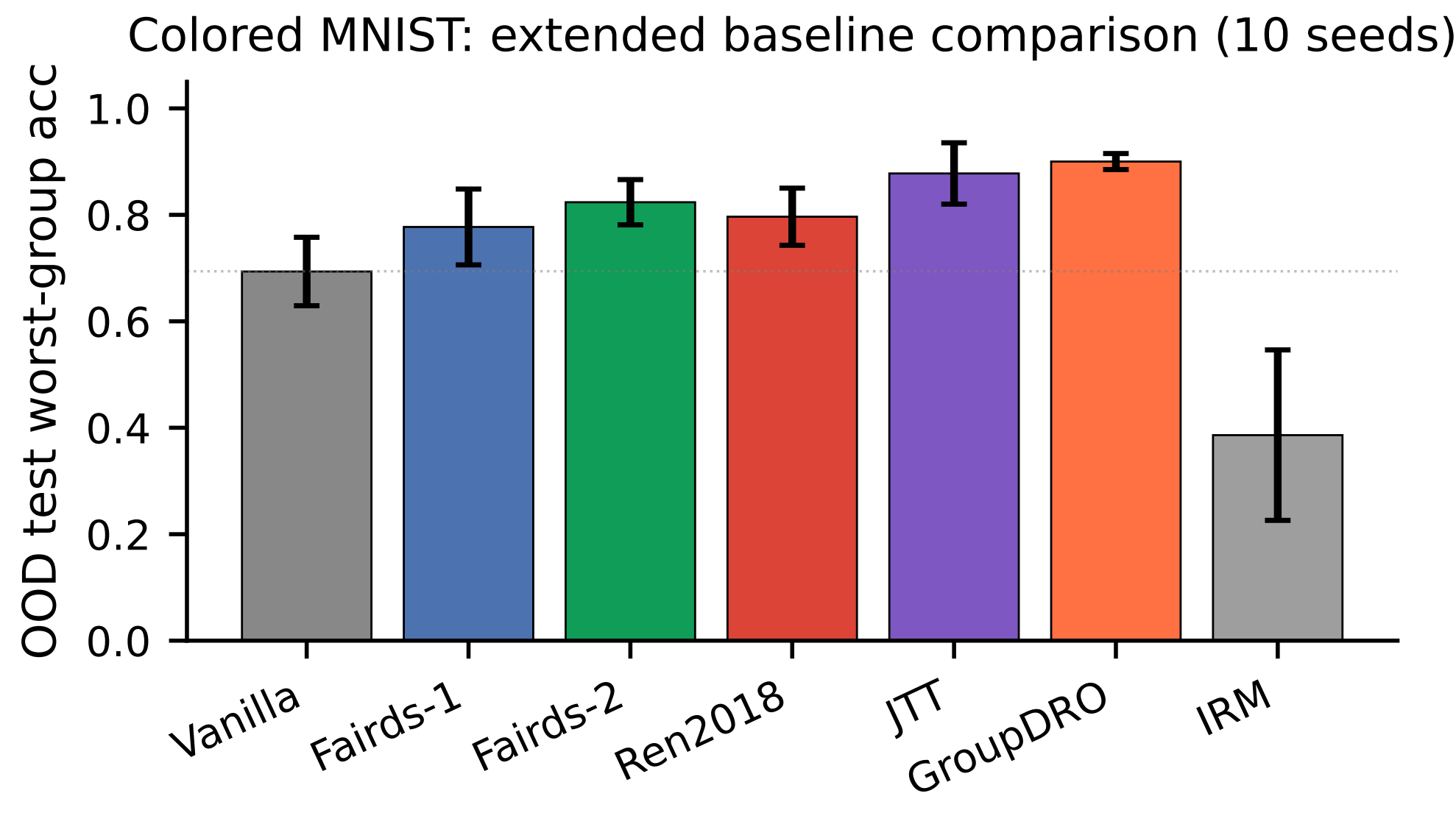
► $H_{\text{val}} g_{\text{val}}$ via Pearlmutter HVP.

► EMA-accumulate ϕ_i , softmax-weight, weighted SGD.

Algorithmic contribution: cross-term must be RMS-rescaled to first-order’s magnitude. Without it Fairds-2 collapses to chance under strong spurious bias.

Colored MNIST (E3): extended baselines

10 seeds $\times$ 7 methods $\times$ 20 epochs, best-tuned $\eta=0.05$.



Held-out OOD test_worst across all baselines.

Fairds-2 vs Vanilla: +13.0 pp ($p=5\times 10^{-5}$).

Fairds-2 vs Ren2018: +2.7 pp ($p=0.052$).

Fairds-2 vs Fairds-1 (isolation): +4.7 pp ($p=0.033$).

Honest: **JTT** (no group label, 2-stage ERM) reaches 0.878 and **GroupDRO** (group label) reaches 0.900, both above Fairds-2’s 0.824.

Mechanism (E1b probe)

Controlled spurious-feature MLP, 2 of 8 dims informative, spurious strength $s=10$.

► Mann-Whitney $U(\text{maj} < \min \phi)$:

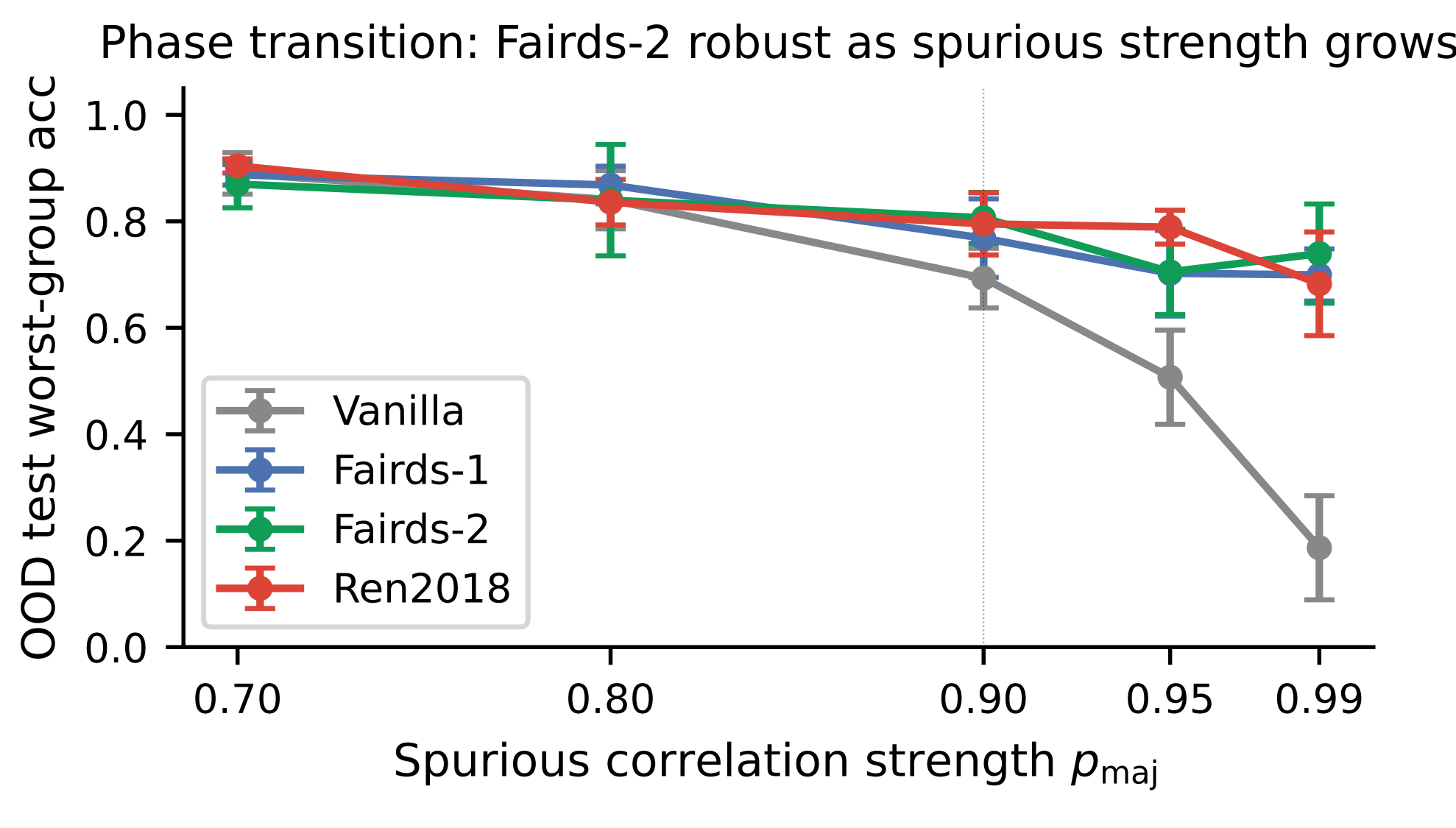
$p=1.89\times 10^{-8}$ at imbalance 0.9.

► $\Delta\phi$ monotone $\downarrow$ with imbalance.

Cross-term penalises majority gradients aligned with H_{val} ’s dominant directions.

Phase transition with spurious strength

Sweep $p_{\text{maj}} \in \{0.7, \dots, 0.99\}$:

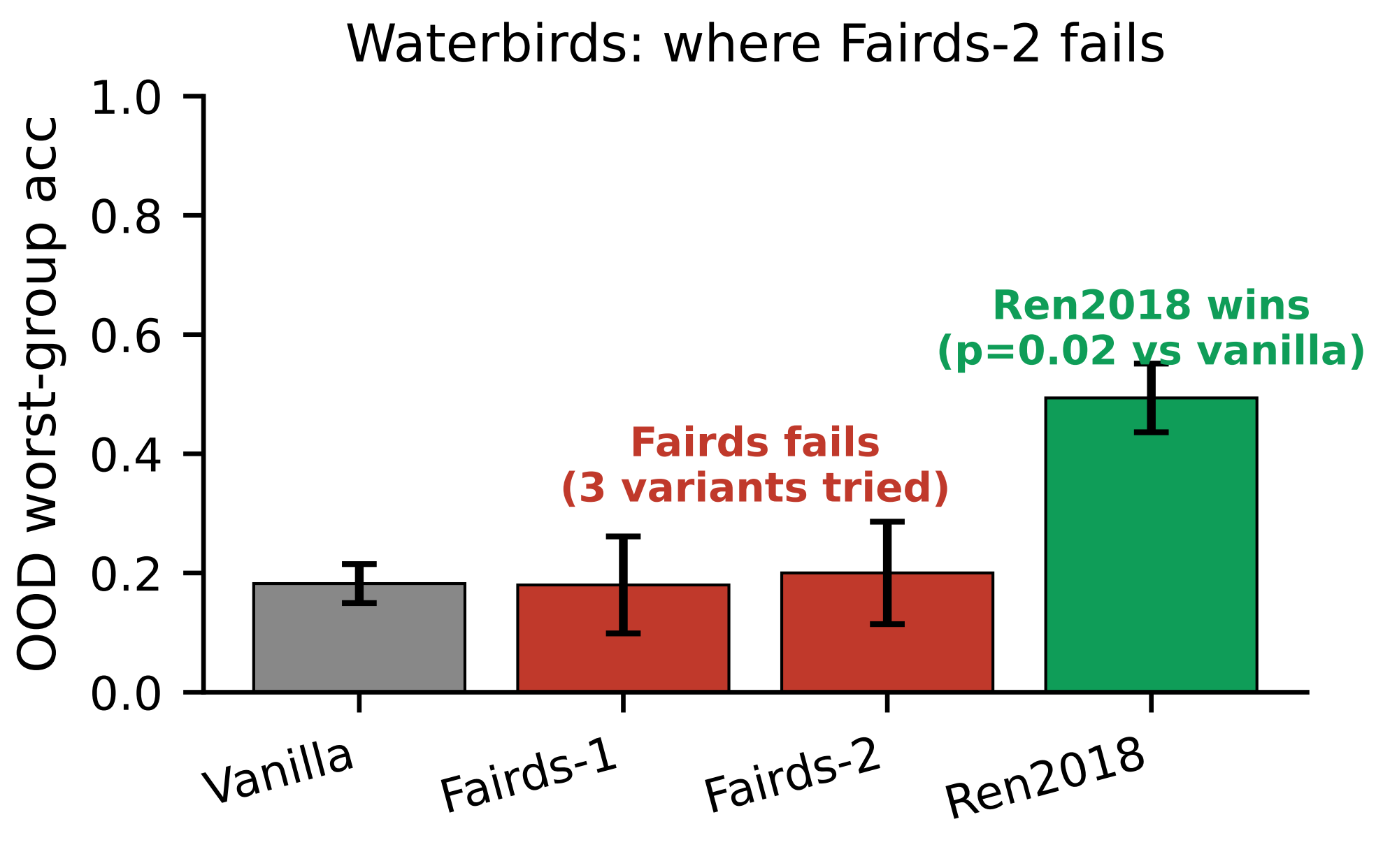


Vanilla collapses at $p=0.99$ while Fairds-2 stays robust.

$p=0.95$: $\Delta=+19.8$ pp; $p=0.99$: $\Delta=+55.3$ pp ($p=3\times 10^{-5}$).

Waterbirds (E3b): regime boundary

Pretrained ResNet-18 fine-tuning. We tried 3 algorithmic variants:



All Fairds variants $\leq$ Vanilla; Ren2018 dominates.

Diagnosis: pretrained g_i is small / noisy in early epochs. EMA fills with noise; reweighting destabilises training. No-EMA variant doesn’t cure it – underlying SNR is poor.

Practical recommendation: **bi-level meta-learning** for pretrained fine-tuning regimes.

Takeaways

- Closed-form 2nd-order Shapley reweighting is a **regime-dependent** tool.
- It works on **from-scratch** / **spurious-correlation** regimes – with statistical isolation of the 2nd-order benefit.
- It **fails on pretrained fine-tuning** – bi-level methods stay stronger there.
- RMS-rescaling of the cross-term is required for stability.